Novelty Analysis

Analysis of Novelty, Transience, and Resonance

PENPAL Project

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Contents

0.1 Summary

- Both AI agents show comparable patterns in novelty (surprisal) and transience metrics.
- Both agents display a positive novelty–resonance relationship (innovation bias), with similar slopes.
- This indicates that novel contributions from both agents have lasting forward influence on narrative development.
- Agent 1 and Agent 2 show similar variance and informational impact in this AI-AI baseline condition.
- The symmetric pattern serves as a baseline for comparison with human-AI and human-human conditions.

1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

1.2 1.2 Reshape Data

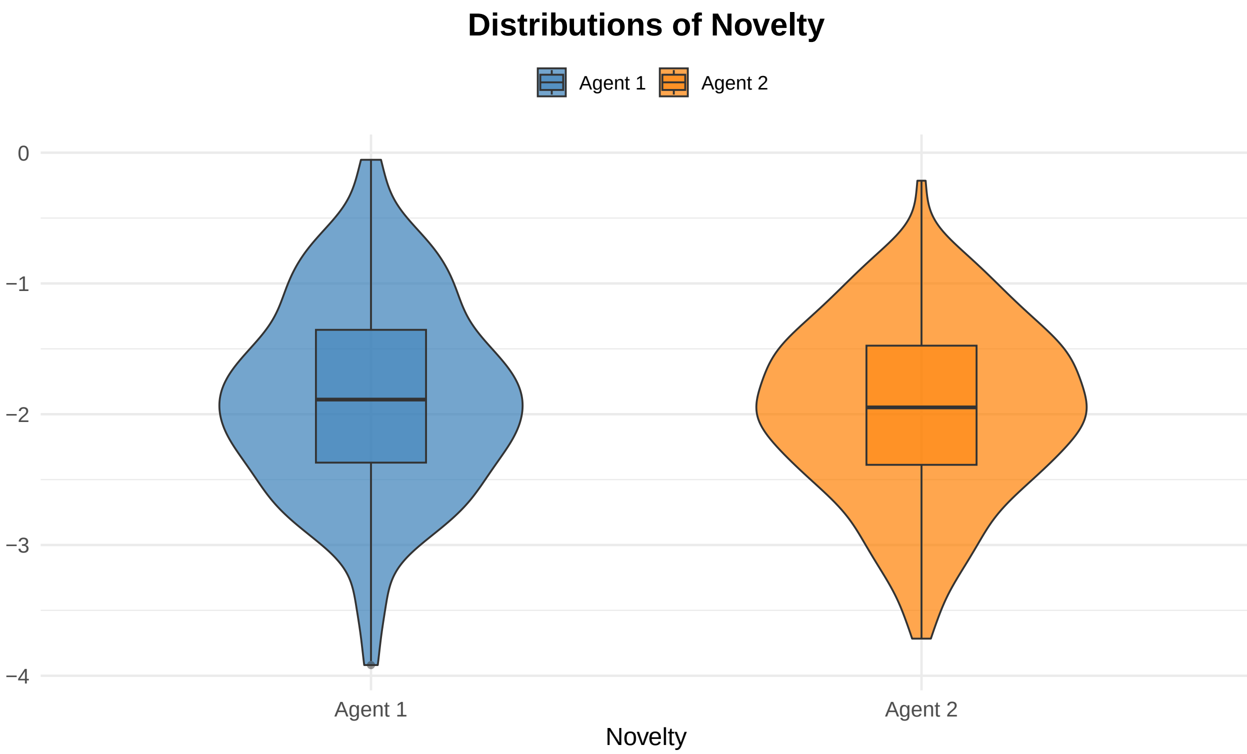
1.3 1.3 Descriptive Statistics

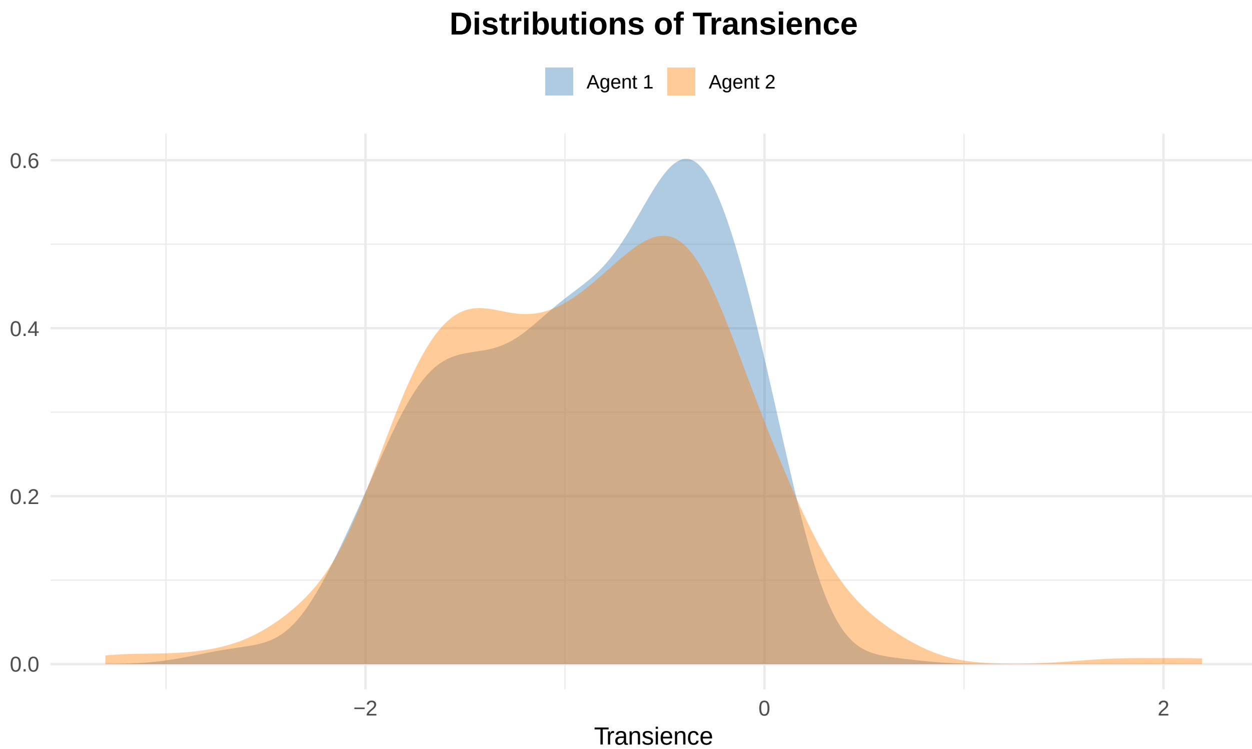
```
## # A tibble: 2 x 8
##   agent       n mean_surprise sd_surprise mean_transience sd_transience
##   <fct>   <int>         <dbl>       <dbl>           <dbl>         <dbl>
## 1 Agent 1   317         -1.87       0.743          -0.881         0.647
```

```
## 2 Agent 2   317         -1.95       0.667          -0.912         0.738
## # i 2 more variables: mean_resonance <dbl>, sd_resonance <dbl>
```

Table 1: Descriptive Statistics for Novelty Metrics by Agent

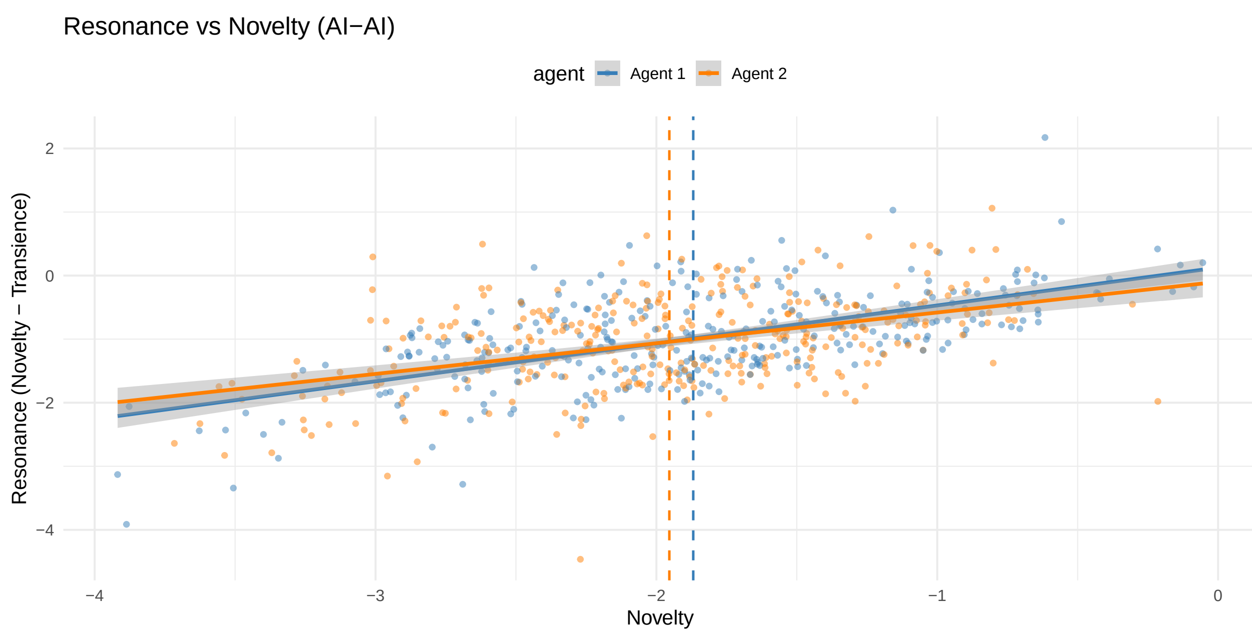
agent	n	mean_surprise	sd_surprise	mean_transience	sd_transience	mean_resonance	sd_resonance
Agent 1	317	-1.869	0.743	-0.881	0.647	-0.988	0.725
Agent 2	317	-1.954	0.667	-0.912	0.738	-1.042	0.728



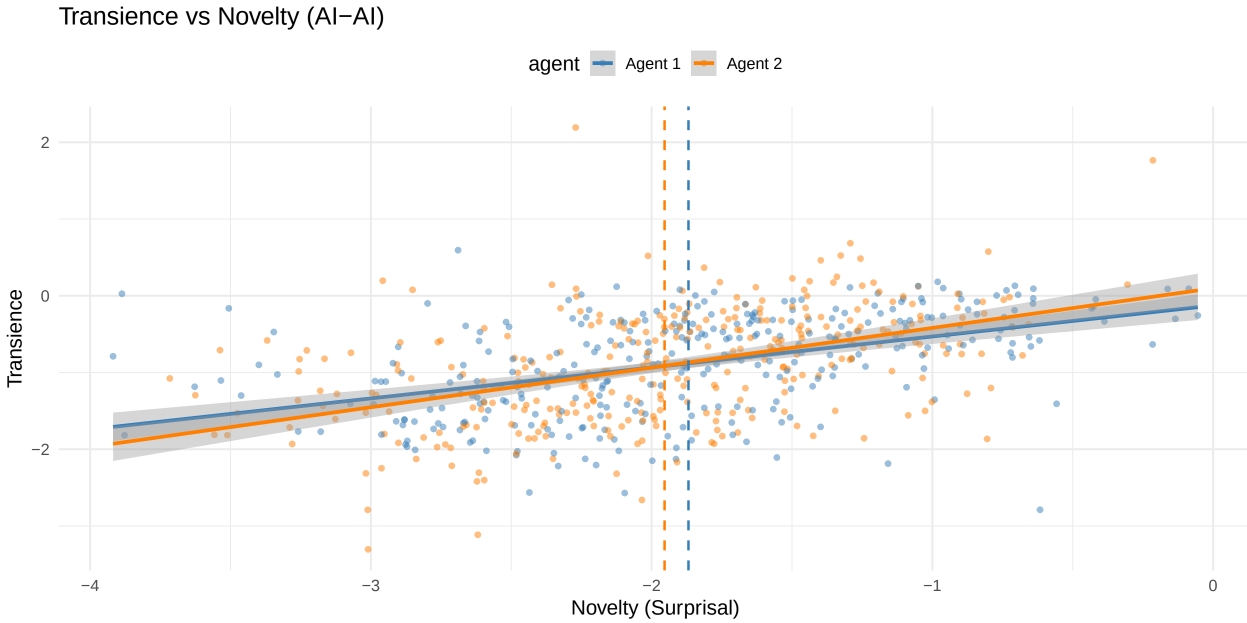


2 2. Novelty and Resonance Analysis

2.1 2.1 Resonance vs Novelty Plot



2.2 2.2 Transience vs Novelty Plot



3 3. Statistical Models

3.1 3.1 Resonance Model

```
## === Resonance Model ===

##                          Estimate Std. Error       df  t value     Pr(>|t|)
## (Intercept)             0.84497609 0.11217173 128.6157  7.532879 7.747987e-12
## surprise                0.97849656 0.04066198 612.6782 24.064161 1.397207e-90
## agentAgent 2           -0.15423686 0.10920545 590.9849 -1.412355 1.583718e-01
## surprise:agentAgent 2 -0.09394164 0.05359688 591.1397 -1.752745 8.016423e-02

##
## === Slope Comparison (Resonance) ===

##  agent   surprise.trend     SE  df lower.CL upper.CL
##  Agent 1          0.978 0.0407 614    0.899    1.058
##  Agent 2          0.885 0.0450 613    0.796    0.973
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

##
## === Resonance Model (controlling for turn length) ===

##                          Estimate  Std. Error       df   t value     Pr(>|t|)
## (Intercept)             0.32889231 0.276471433 597.4673  1.189607 2.346735e-01
## surprise                0.96913017 0.040792447 612.5731 23.757588 6.293622e-89
## agentAgent 2           -0.14692318 0.109094924 589.8500 -1.346746 1.785792e-01
## nwords                  0.01810899 0.008915812 621.8479  2.031110 4.266867e-02
## surprise:agentAgent 2 -0.08742236 0.053609713 590.0853 -1.630719 1.034833e-01

##
## === Slope Comparison (Resonance, length-controlled) ===
```

```
##  agent   surprise.trend     SE  df lower.CL upper.CL
##  Agent 1          0.969 0.0408 614    0.889     1.05
##  Agent 2          0.882 0.0450 613    0.793     0.97
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

3.2 3.2 Transience Model

```
## === Transience Model ===
```

```
##                         Estimate Std. Error       df   t value     Pr(>|t|)
## (Intercept)          -0.84497609 0.11217173 128.6157 -7.532879 7.747988e-12
## surprise              0.02150344 0.04066198 612.6782  0.528834 5.971121e-01
## agentAgent 2          0.15423686 0.10920545 590.9849  1.412355 1.583718e-01
## surprise:agentAgent 2  0.09394164 0.05359688 591.1397  1.752745 8.016423e-02
```

```
##
## === Slope Comparison (Transience) ===
```

```
##  agent   surprise.trend     SE  df lower.CL upper.CL
##  Agent 1         0.0215 0.0407 614  -0.0584    0.101
##  Agent 2         0.1154 0.0450 613   0.0270    0.204
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
##
## === Transience Model (controlling for turn length) ===
```

```
##                         Estimate  Std. Error       df    t value   Pr(>|t|)
## (Intercept)          -0.32889231 0.276471433 597.4673 -1.1896069 0.23467347
## surprise              0.03086983 0.040792447 612.5731  0.7567536 0.44948843
## agentAgent 2          0.14692318 0.109094924 589.8501  1.3467463 0.17857918
## nwords               -0.01810899 0.008915812 621.8479 -2.0311101 0.04266867
## surprise:agentAgent 2  0.08742236 0.053609713 590.0853  1.6307186 0.10348329
```

```
##
## === Slope Comparison (Transience, length-controlled) ===
```

```
##  agent   surprise.trend     SE  df lower.CL upper.CL
##  Agent 1         0.0309 0.0408 614  -0.0493    0.111
##  Agent 2         0.1183 0.0450 613   0.0300    0.207
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

---

4 5. Session Information

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS 26.3
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
```

```
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK ve
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Australia/Melbourne
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_1.10.6  here_1.0.1      afex_1.4-1      cowplot_1.1.3
##  [5] patchwork_1.3.0 zoo_1.8-12      stringi_1.8.4   arrow_19.0.1.1
##  [9] lmerTest_3.1-3  lme4_1.1-35.5   Matrix_1.7-1    lubridate_1.9.3
## [13] forcats_1.0.0   stringr_1.5.1   dplyr_1.1.4     purrr_1.0.2
## [17] readr_2.1.5     tidyr_1.3.1     tibble_3.2.1    ggplot2_3.5.1
## [21] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
##  [1] tidyselect_1.2.1    farver_2.1.2        fastmap_1.2.0
##  [4] TH.data_1.1-2       pacman_0.5.1        digest_0.6.37
##  [7] timechange_0.3.0    estimability_1.5.1  lifecycle_1.0.4
## [10] survival_3.7-0      magrittr_2.0.3      compiler_4.4.2
## [13] rlang_1.1.4         tools_4.4.2         utf8_1.2.4
## [16] yaml_2.3.10         knitr_1.50          labeling_0.4.3
## [19] bit_4.0.5           plyr_1.8.9          abind_1.4-8
## [22] multcomp_1.4-26     withr_3.0.2         numDeriv_2016.8-1.1
## [25] grid_4.4.2          fansi_1.0.6         xtable_1.8-4
## [28] colorspace_2.1-1    scales_1.3.0        MASS_7.3-61
## [31] tinytex_0.52        cli_3.6.3           mvtnorm_1.3-3
## [34] rmarkdown_2.28      crayon_1.5.3        ragg_1.3.2
## [37] generics_0.1.3      rstudioapi_0.16.0   reshape2_1.4.4
## [40] tzdb_0.4.0          minqa_1.2.8         splines_4.4.2
## [43] assertthat_0.2.1    parallel_4.4.2      vctrs_0.6.5
## [46] boot_1.3-31         sandwich_3.1-1      carData_3.0-5
## [49] car_3.1-3           hms_1.1.3           pbkrtest_0.5.3
## [52] bit64_4.0.5         Formula_1.2-5       systemfonts_1.1.0
## [55] glue_1.7.0          nloptr_2.1.1        codetools_0.2-20
## [58] gtable_0.3.5        munsell_0.5.1       pillar_1.9.0
## [61] htmltools_0.5.8.1   R6_2.5.1            textshaping_0.4.0
## [64] rprojroot_2.0.4     vroom_1.6.5         evaluate_1.0.3
## [67] lattice_0.22-6      backports_1.5.0     broom_1.0.6
## [70] Rcpp_1.0.13         coda_0.19-4.1       nlme_3.1-166
## [73] mgcv_1.9-1          xfun_0.56           pkgconfig_2.0.3
```

Analysis complete. All figures saved to `analysis/figures/` and tables to `analysis/tables/`.